Supplementary Information for

Synonymous and non-synonymous codon substitutions can alleviate dependence on GroEL for folding

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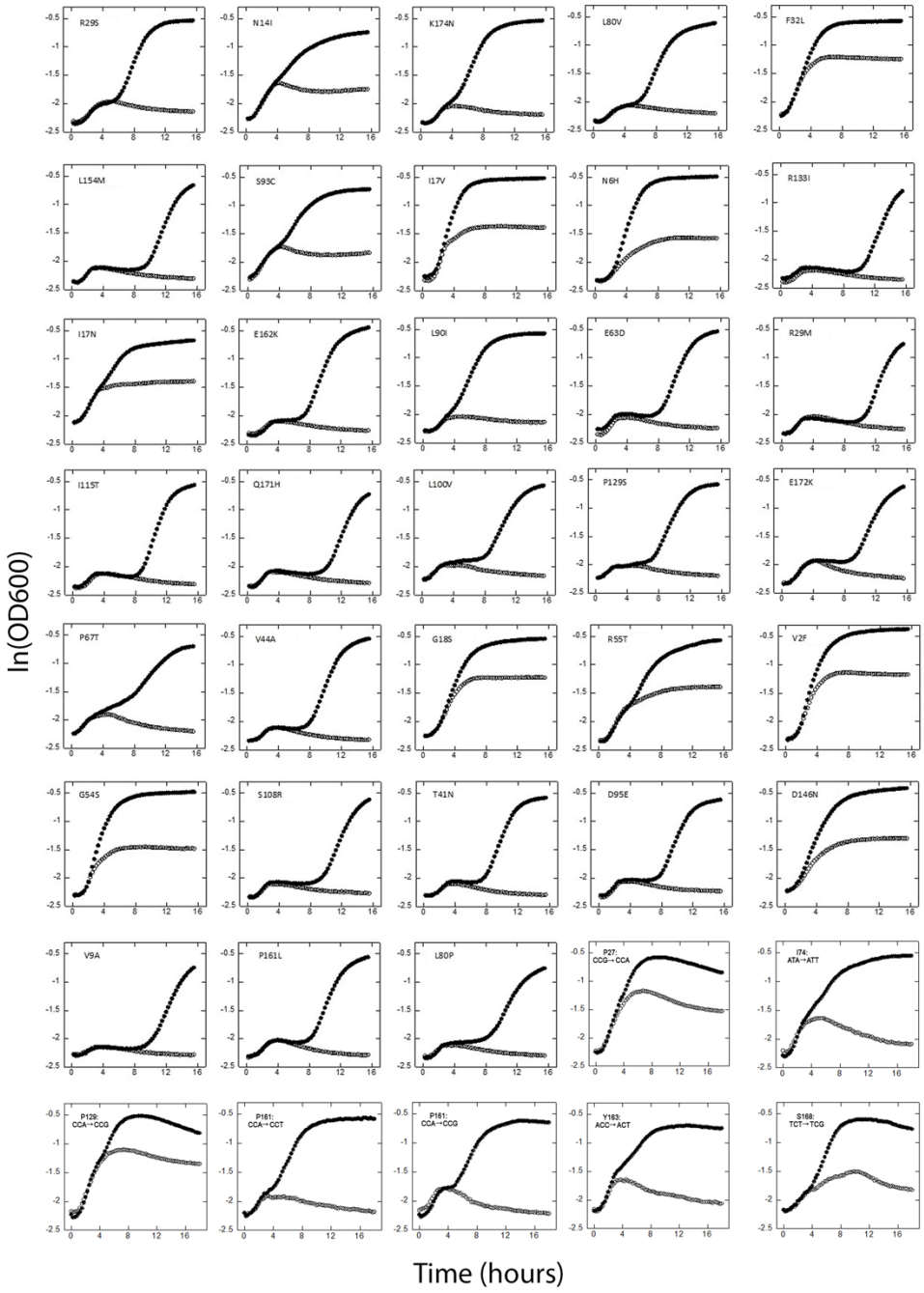


Figure S1. Growth curves of mDHFR variants in the presence of arabinose vs. glucose. Clones expressing mDHFR variants which appeared to support *E. coli* growth on plates, upon GroEL depletion, were sequenced and grown in liquid medium containing antibiotics, TMP, DAP and either glucose or arabinose. It may be seen that cells expressing the mDHFR mutants G18S or F32L mutants, for example, grow better in liquid medium containing glucose vs. arabinose, compared to cells expressing wild-type mDHFR.

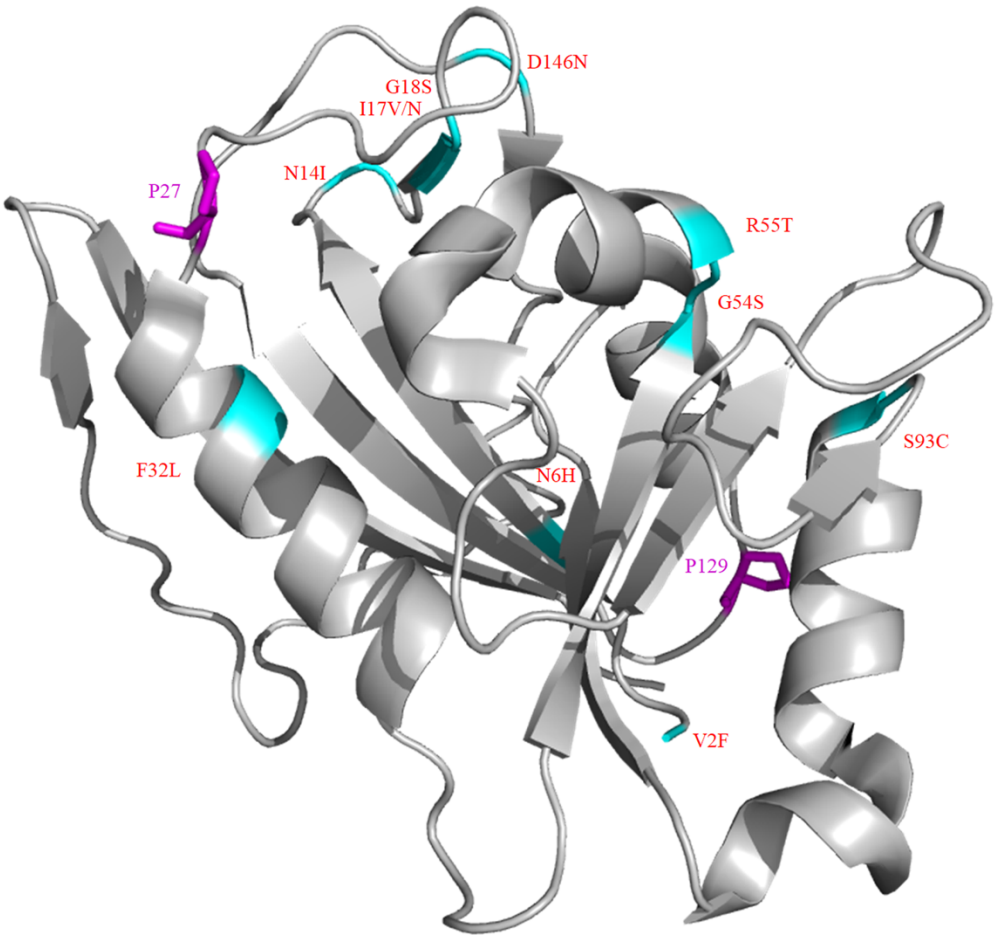


Figure S2. Ribbon diagram of the crystal structure of mDHFR (PDB ID: 3d84). The positions of the missense mutations in the protein backbone are shown in cyan and the mutations are indicated in red using the single-letter code for amino acids. The proline residues corresponding to the silent mutation sites are shown in purple. The figure was generated using PyMOL.

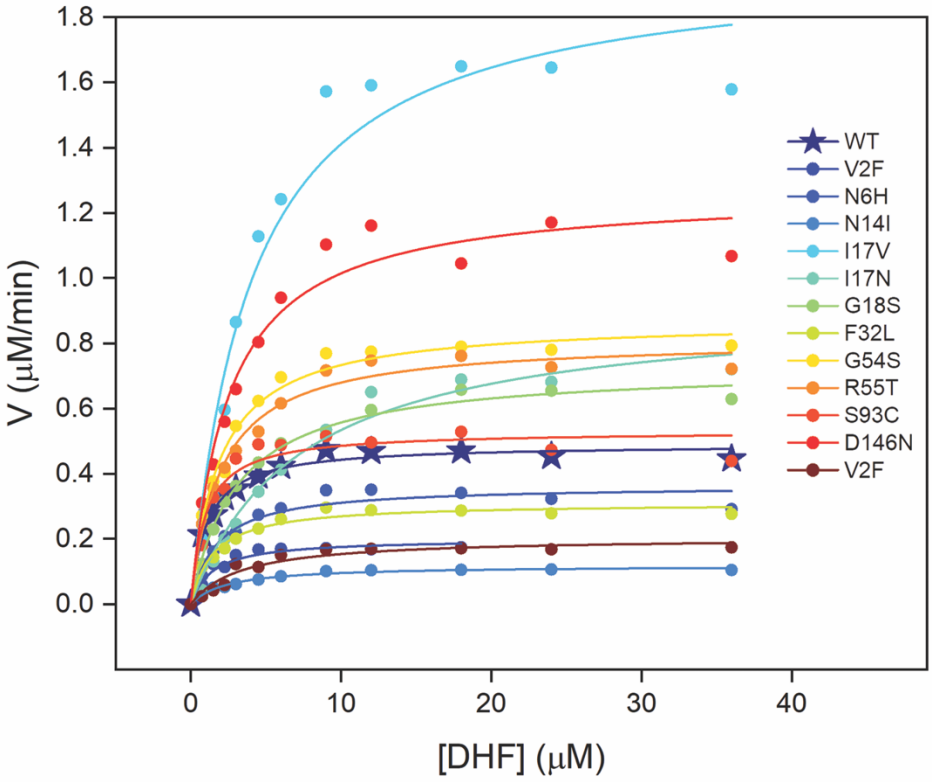


Figure S3. Steady-state kinetics of dihydrofolic acid (DHF) reduction by the mDHFR variants**.** Shown are representative plots of initial rates of DHF reduction by the mDHFR variants as a function of dihydrofolate concentration. The initial rates were obtained from the linear phase of the reactivation curves and the data were fitted to the Michaelis-Menten equation.

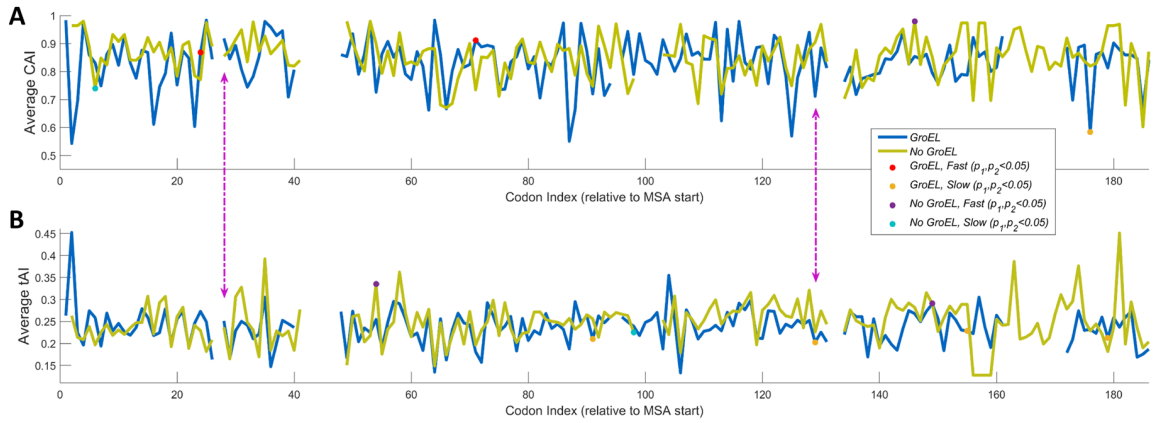


Figure S4. Bioinformatic analyses of the nucleotide sequences of DHFR in mollicutes with or without GroEL. Shown are the average values of CAI (A) and tAI (B) for the organisms with GroEL (GroEL$^+$, blue plots) and without GroEL (GroEL$^-$, olive yellow plots). The two silent mutations are marked by purple arrows. The inset shows the color coding for dots that indicate positions in GroEL$^+$ sequences with high levels of selection for fast (red) and slow (orange) translated codons, and positions in GroEL$^-$ sequences with high levels of selection for fast (purple) and slow (blue) translated codons. The two statistical tests used to find significant points (the p-values p_1 and p_2) are described in the Methods.

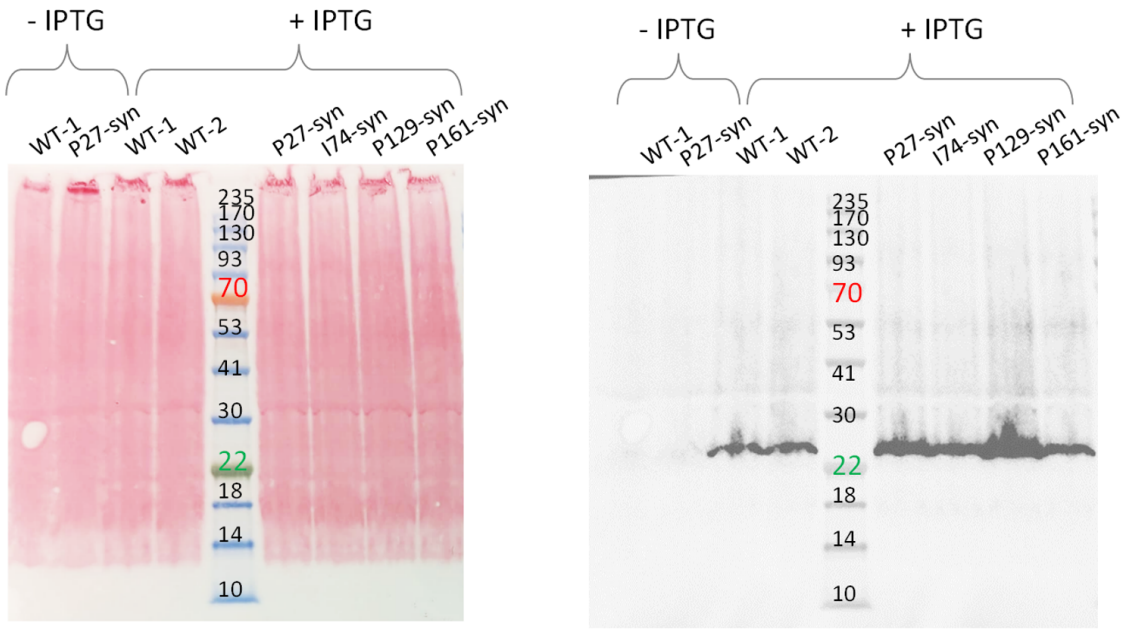


Figure S5. Effects of synonymous mutations on the levels of expression of mDHFR. Overnight cultures of MGM100 cells expressing the appropriate mutant and containing 0.2% arabinose, 100 μg/ml ampicillin and 30 μg/ml kanamycin were diluted 1:100, grown to OD_{600} of 0.6 at 37 °C and expression was then induced by adding IPTG. Cells were then grown for another 4 hours at 37 °C, pelleted and samples were then prepared for Western blotting. The ponceau red staining on the left shows that the samples in the different lanes correspond to similar numbers of cells. Western blotting of these samples using an anti-His tag antibody (right panel) was employed to determine the expression levels of the different mutants. The slightly higher levels of expression of the proteins with the two synonymous mutations at positions 27 and 129, relative to wild-type mDHFR, is consistent with codon usage improving protein folding yields. Synonymous mutations at positions 74 and 161, which do not affect GroEL dependence, also lead to somewhat higher expression levels, thus indicating that release from GroEL dependence is not a simple function of expression level.